



BHABHA UNIVERSITY,BHOPAL(M.P)

PROGRAME NAME.... DIPLOMA MECHANICAL ENGINEERING

SCHEME OF STUDIES AND EXAMINATION FOR : SIXTH SEMESTER

Session:2021

COURSE CODE	COURSE TITLE	THEORY COMPONENT												TOTAL CREDIT	GRAND TOTAL MARKS
		LECTURE	CONTINEOUS EVALUATION		END OF THE TERM/SEMESTER EVALUATION		THEORY CREDIT	PRACTICAL PER HOURS	CONTINEOUS EVALUATION	END OF THE SEMESTER		PRACTICAL CREDIT			
			Hrs PER WEEK	TERM WORK, QUIZ, ASSIGNMENT	MID TERM					THEORY PAPER			LAB WORK, QUIZ, ASSIGNMENT		
		I	II	MARKS	DURATION(Hrs)		MARKS	DURATION							
ME-6001	DESIGN OF MACHINE ELEMENTS	4	10	10	10	70	3	4						4	100
ME-6002	AUTOMOBILE ENGINEERING	4	10	10	10	70	3	4	4	20	30	3	2	6	150
ME-6003	POWER PLANT ENGINEERING	4	10	10	10	70	3	4	4	20	30	3	2	6	150
ME-6004	MAJOR PROJECT								14	100	200		14	14	300
ME-6005	PROFESSIONAL ACTIVITY								2						
TOTAL		12	30	30	30	210	9	12	24	140	260		18	30	700



BHABHA UNIVERSITY, BHOPAL (M.P.)

DIPLOMA IN MECHANICAL ENGINEERING

ME-6001 DESIGN OF MACHINE ELEMENTS

SIXTH SEMESTER

SCHEME OF STUDIES

S. No.	Topics	Theory Hrs	Practical Hrs	Total Hrs
1	Introduction to M/c design.	06	-	06
2	Design of M/c elements subjected to direct and shear loads.	12	-	12
3	Design of M/c. elements subjected to bending and twisting and combined bending and twisting.	12	-	12
4	Design of riveted joint	04	-	04
5	Design of Simple welded joints.	04	-	04
6	Design of threaded joints.	06	-	06
7	Design of clutch	06	-	06
8	Selection of Rolling and Sliding contact bearing	10	-	10
	Total	60		60



BHABHA UNIVERSITY, BHOPAL (M.P.)

DIPLOMA IN MECHANICAL ENGINEERING

ME-6001 DESIGN OF MACHINE ELEMENTS

SIXTH SEMESTER

Lectures: 04 Hrs. per Week

S.N O	CONTENT	STU DY Hrs.
1	Introduction to Machine Design: Machine and machine elements, bolt, nut, axle, shaft, bearing, coupling, clutch, belt, rope, chain, gear etc. Specific purpose of piston connecting rod, crank shaft, turbine blade etc. Factors influencing design of machine elements – Strength stiffness, light weight, wear resistance minimum size, availability, processability, safety, compliance with standards. Basic design procedure. Selection of mechanism, material, shape and size. Preliminary design, applying checks, revision of design final design. Factors influencing selection of materials. Type of failures, types of forces. Types of loading. Safe design stress and factors of safety.	06
2	Design of Machine Elements Subjected to Direct and Shear Loads: Introduction members subjected to direct loads – bolt, column, rod, cotter and knuckle joints, members subjected to shear loads rivet, cotter knuckle pin, root of threaded bolt, coupling, bolt, key. Function, application and design of knuckle and cotter joint.	12
3	Design of Machine Elements Subjected to Bending Moment, Twisting Moment and Combined Bending and Twisting Moment : Introduction to pure bending, fundamental equation of pure bending viz : $M/I = f/y = E/R$ Design of shaft, key, flange coupling, leaf and helical spring, pulley arms, axle	12



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DIPLOMA IN MECHANICAL ENGINEERING

ME-6001 DESIGN OF MACHINE ELEMENTS

SIXTH SEMESTER

S.N O	CONTENT	STUD Y Hrs
4	Design of Riveted Joint: Type of fastening - temporary and permanent, types of riveted joint - lap and butt joint, definition of common terms like pitch, back pitch, efficiency, margin. Modes of failure of riveted joints	12
5	Design of Simple Welded Joints: Definition of welding advantages of welding over riveted joints, types of welded joints, strength of the butt weld, types of fillet joints and strength of fillet joint problem solving.	04
6	Design of Threaded Joints: Types of threads and their proportions, Types of bolts, proportion of nut and bolt dimensions, design of bolt, designation of threads as per I.S. codes.	06
7	Design of Clutch: Pivots and Collars friction. Horse power lost assuming uniform pressure and uniform wear. Clutch-need, classification and construction and working of single and multi plate clutches, horse power transmitted by single and multi plate clutches.	06
8	Selection of Rolling and Sliding contact Bearing: Types of rolling contact bearing , Ball bearing Roller bearing, bearing designation, bearing installation. Application of bearing. Basic principle of Hydro dynamic and Hydro static bearing. Bearing modulus and Bearing characteristics number.	10



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DIPLOMA IN MECHANICAL ENGINEERING

ME-6001 DESIGN OF MACHINE ELEMENTS

SIXTH SEMESTER

REFERENCE BOOKS

1. Machine Design by Sharma and Agrawal.
2. Machine Design by R.K. Jain.
3. Machine Design by Shigley..
4. Machine Design by R S Khurmi
5. Introduction to Machine Design by Bhandari Tata Mcgraw Hill
6. Machine Design by Pandya and Shah



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DIPLOMA IN MECHANICAL ENGINEERING

ME-6002 AUTOMOBILE ENGINEERING

SIXTH SEMESTER

SCHEME OF STUDIES

Lectures: 04 Hrs. per Week

Practical: 04 Hrs. per Week

Lectures: 03 Hrs. per Week Practical: 04 Hrs. per Week

S.No	Topics	Theor y Hrs	Practic al Hrs	Total Hrs.
1	Introduction	0 3	0 3	0 6
2	Auto engines.	1 6	0 9	2 5
3	Auto electrical system	1 0	0 9	1 9
4	Transmission system.	0 6	0 9	1 5
5	Final drive system	0 4	0 6	1 0
6	Braking system.	0 6	0 6	1 2
7	Front axle and steering.	0 4	0 6	1 0
8	Frame and suspension	0 4	0 6	1 0
9	Miscellaneous	0 7	0 6	1 3
	Total	60	60	120



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DIPLOMA IN MECHANICAL ENGINEERING

ME-6002 AUTOMOBILE ENGINEERING

SIXTH SEMESTER

Lectures: 03 Hrs. per Week

S.N O	CONTENT	STU DY Hrs.
1	Introduction: Meaning of automobile, elements of automobile, classification of automobile, layout of chassis, various operating systems used in automobile.	03
2	Auto Engines: Meaning of I.C. Engines, Classification on the basis of cycle, fuel used, ignition system, number of cylinders, number of strokes etc. Otto/ Diesel cycles. Two stroke and four stroke engines, merits and demerits, scavenging comparison of petrol and diesel engines. Cooling systems, firing order. Valve timing diagrams. Engine rating. Lubrication, factors affecting lubrication, Lubrication systems, Fuel Supply system, fuel pump - SPU electric pump. Carburettor, air fuel ratio, Solex and amal carburettor.	16
3	Auto Electric System: Wiring diagram of a car and functions of various components used in the electric circuits, function and working principle of a starter and generator, function of voltage- current regulator, ignition timing, spark plugs- their classification, gap setting and common ignition troubles, their causes and remedies . Automobile battery - construction and working, electronic ignition system of modern vehicles.	10
4	Transmission System: Clutch : necessity, function of its components, Types –single & multi plate and centrifugal clutches, clutch actuating mechanism and fluid flywheel. Gear Boxes : necessity, Types of gear boxes and their working. Importance of gear shifting mechanism, gear box troubles, their causes and remedies.	6



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DIPLOMA IN MECHANICAL ENGINEERING

ME-6002 AUTOMOBILE ENGINEERING

SIXTH SEMESTER

S.N O	CONTENT	STU DY Hrs.
5	Final Drives System: Drive mechanism in cars, purpose and working of propeller shaft, construction of propeller shaft. Types of universal joints. Rear axle assembly : function of differential - constructional features and working. Arrangement of semi floating and fully floating rear axle, and their troubles.	04
6	Braking system: Introduction, classification of brakes, construction & working of mechanical brake, hydraulic brake, Electric brake , advantages and disadvantages of each type of brakes, Servo brake system.	06
7	Front Axle and Steering: Function of front axle, axle type, wheel alignment and its elements toe- in, toe -out. King pin inclination. Ackerman steering principle. Camber and castor angle . Elements of steering - types and working ,Under and over steering, power steering and advanced steering systems.	04
8	Frame and Suspension: Frames : necessity, function, Classification, suspension system, types, leaf, coil spring. Telescopic shock absorber. Air suspension, independent suspension system. Tyres : structure of tyre section, rating of tyres, tyre- pressure measurement, material and specification. Tyre wear and remedies.	04
9	Miscellaneous: (i) Maintenance of Vehicles : need, classification maintenance procedure of engine, transmission system, electrical system,braking system and steering mechanism. (ii) Garage and Service Station: Types, layout, equipment tools and service procedure. (iii)Exhaust Gas, Pollutants: Their hazards and controls with reference to motor vehicle act. Motor Vehicle act, registration of vehicles, driving license and Traffic Signals.	07



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DIPLOMA IN MECHANICAL ENGINEERING

ME-6002 AUTOMOBILE ENGINEERING

SIXTH SEMESTER

LIST OF EXPERIMENTS

S.No.	EXPERIMENT	PRACT. Hrs.
1	Study of automobile chassis and function of various components.	04
2	Study of various components used in auto vehicles. (a) Engine (b) Transmission system (c) Final Drive (d) Braking system (e) Electrical system (f) Steering Mechanism	08
3	Study of various elements of scooter/ car/bus servicing.	06
4	Engine tuning and adjustment for smooth, idle speed of a scooter/moped/ motor cycle/ car etc.	06
5	Starting troubles and their rectifications.	04
6	Rectification procedure for : (a) Air bleeding. (b) Brake adjustment. (c) Problems in Carburettor. (d) Wheel alignment.	06
7	Battery, servicing and charging.	06
8	Repair of punctured tyre and re-treading of tyres.	04
9	Study of auto servicing centre. (a) Layout (b) Instruments/ Tools used (c) Servicing procedures.	06
10	Visit of a local auto service centre and prepare a report in respect of : (a) Layout (b) Instruments/ Tools used (c) Servicing/ Reconditioning/ Maintenance procedure.	06
11	Collect specifications for 2/3/4 Wheeler and prepare a comparison table and their manuals.	06



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DIPLOMA IN MECHANICAL ENGINEERING

ME-6002 AUTOMOBILE ENGINEERING

SIXTH SEMESTER

REFERENCE BOOKS

- 1 Automobiles Engineering Vol. I & I I by Dr. Kirpal Singh. (Standard Publisher)
- 2 Automobiles Engineering by R.S. Gupta (Satya Prakashan)
- 3 Automobile machanism by Joseph Heither
- 4 Automobile Engineering by R. P. Sharma (Dhanpat Rai & Sons)
- 5 Automobile Mechanism by William H. Crouse
- 6 I.C. Engines by Dr. A.C. Rad and S.B. Bechar
- 7 Automobile Engineering- T.R. Banga & Nathu Singh (Khanna Publicers)
- 8 Automobile Engg. – RB. Gupta
- 9 Automobile Engg. – K.M. Agrawal (Vol. I & II)
- 10 Automobile Engineering by Prof. S.M. Pande and K.K. Jain (Deepak Prakashan, Morar Gwalior)
- 11 IS: 202-1602-1960 Code for testing of variable speed I.C. Engines for automobile purpose.
- 12 IS: 1603-1960 Performance of variable speed I.C. Engines for automobile purpose.
- 13 IS : 1634-1964 Helical springs for automobile suspension.
- 14 IS : 2742-1964 Automobile brake lining.
- 15 IS : 3511-1966 Cylinder bore diameters for I.C. Engine.
- 16 IS : 4552-1963 Portable Jacks for automobile, Mechanical and hydraulic.
- 17 IS : 5423-1969 General requirements for hydraulic shock absorber for automobile suspension, direct acting telescope type.
- 18 IS : 5791-1970 Technical supply conditions for piston ring for IS Engines.
- 19 IS : 4974-1963 Grease nipples, small



BHABHA UNIVERSITY, BHOPAL (M.P.)

DIPLOMA IN MECHANICAL ENGINEERING

ME-6003 POWER PLANT ENGINEERING

SIXTH SEMESTER

SCHEME OF STUDIES

Lectures: 04 Hrs. per
Week

Practical: 04 Hrs. per
Week

S No.	Topic s	Theory Hrs.	Practical Hrs	Tota l Hrs.
1	Steam power plant	08	08	16
2	Steam generators	08	08	16
3	Steam prime-mover	07	08	15
4	Condensing unit	04	06	10
5	Steam power station control and safety	06	06	12
6	Nuclear power station.	08	04	12
7	Diesel engine power station	08	08	16
8	Gas turbine power plant.	08	08	16
9	Hydroelectric plant.	03	04	07
Total		60	60	120



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DIPLOMA IN MECHANICAL ENGINEERING

ME-6003 POWER PLANT ENGINEERING

SIXTH SEMESTER

S.NO	CONTENT	STUDY Hrs.
1	Steam Power Plant: Energy conversion in a thermal power station. Limitations on conversion of heat into work, direct conversion devices, central power station, industrial power station, captive power station, advantages. Classification of power station on the basis of prime-movers. Elements of steam power plant, function of each element- generating unit, prime mover, auxiliary equipment and turbo generator. Revision & Improvement of thermal efficiency of Rankine cycle by lowering exhaust pressure, increasing boiler pressure and superheating of steam. Simple problems on Rankine efficiency. Reheat cycle: Representation on T-S and H-S planes, flow diagram and advantages. Simple regenerative cycle: flow diagram, representation on T-S and H-S planes, bleeding and feed water heating and pumping.	08
2	Steam Generators: Classification according to working pressure Accessories - Super heater, economizer, pre-heater and draft equipment, superheat control methods, pulverized fuel- necessity, storing system. High pressure boiler in modern steam power plant need, features and function of Velox, Benson, Lamaunt, Leoffler high pressure boiler.	08



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ME-6003 POWER PLANT ENGINEERING

SIXTH SEMESTER

S.NO	CONTENT	STUDY Hrs.
3	Steam Primemover: Steam nozzle-Types, velocity of steam at outlet, weight of discharge, area of cross- section at throat and outlet, critical pressure ratio, nozzle efficiency, concept of primemover, steam turbine- Revision of steam turbine in terms of principle of working, methods of compounding and governing, losses in steam turbines, lubrication system of steam turbines.	07
4	Condensing Unit: Steam Condenser, functions, type-jet and surface. Limitations and advantages, elements of condensing unit-cooling towers. .	04
5	Steam Power Station Control and Safety: Effect of load variation on shaft speed, steam admission, valve opening, steam flow rate, steam pressure and combustion control system. Necessity of controlling factors in load variation, control system (area system, centralized control system) functions of annunciator panel system, basic elements of control system, controls and instruments located in a modern central station. Control room, records and their purpose, log sheets or log book. .	06



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ME-6003 POWER PLANT ENGINEERING

SIXTH SEMESTER

S.NO	CONTENT	STUDY Hrs.
6	Nuclear Power Stations: Nuclear reactions - fission, fusion, mass defect, binding energy, chain reaction, types of nuclear materials - fissile materials, fertile materials, process of conversion of fertile materials, breeding moderation. Nuclear reactor - Function- elements of a nuclear reactor- Reacted core, moderator, thermal - Shielding reflector, reactor vessel, fuel, coolant flow, control roads, biological shielding coolants (Caseansnon-boiling liquid, boiling liquid) Fluids - Helium, Co₂ , O₂ under pressure, pressurised water (Ordinary, heavy) liquid metals (Li, Si, Pb, Na) and their alloys, boiling water.Operation - Fast reactors, thermal reactor, breeding reactor. Nuclear fuel - Heterogeneous, Homogeneous. Moderator: Water moderator, heavy water moderator, graphite moderator and Berylling moderator. Health hazards in nuclear power station- Unit of radiation safe and dangerous zones of radiations, safety precautions in a nuclear power station- Medical requirements. Entry requirements: In contamination zones, precaution during changing of fuel. Nuclear waste disposal.	08



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7	Diesel Power Plants: Advantages and disadvantages as a primemover for power generation, essential components of diesel power plant and function. Cooling and lubrication system, fuel injection system- Basic requirements, fuel injection system -common rail system, individual pump system, distribution system, data recording, performance.	08
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DIPLOMA IN MECHANICAL ENGINEERING

ME-6003 POWER PLANT ENGINEERING

SIXTH SEMESTER

S.NO	CONTENT	STUDY Hrs.
8	Gas Turbine Powers Plants: Advantages of gas turbines over I.C. Engine as prime movers, Brayton or Joule cycle, schematic diagrams for open and closed cycles, representation of cycle on P.V. and T.S. diagram. Thermal efficiency in terms of terminal temperature and pressure, effect of pressure ratio on thermal efficiency, advantages and disadvantages of open and closed cycle gas turbines, important components of a gas turbine power plant, methods of improving thermal efficiency, essential auxiliaries and controls of a gas turbine power plant, fuels for gas turbines.	08
9	Hydro Electric Plants: Types, Comparison of low, medium and high heat plants, elements of hydro power plants, governing of turbines, performance of water turbines, site selection.	03



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DIPLOMA IN MECHANICAL ENGINEERING

ME-6003 POWER PLANT ENGINEERING

SIXTH SEMESTER

LIST OF EXPERIMENTS

S.No.	EXPERIMENT	PRACT. Hrs.
1	Study of constructional and working details of : (a) Simple steam power plant. (b) Nuclear power plant. (c) Gas power plant. (d) Diesel power plant. (e) Hydro electric plant.	30
2	Study of controls provided in power plants listed above.	10
3	Draw balance sheet for diesel engine.	05
4	Performance and characteristics of steam/ hydraulic turbines.	05
5	Visits of various power plants located in Madhya Pradesh and submit report in terms of : (a) Detailed layout (b) Capacity. (c) Elements of each unit. (d) Control systems provided. (e) Berometers which are being controlled. (f) Doses occuring in various units. (g) Maintenance of schedule. (h) Lubrication systems, uses.	10



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DIPLOMA IN MECHANICAL ENGINEERING

ME-6003 POWER PLANT ENGINEERING

SIXTH SEMESTER

REFERENCE BOOKS

1. Course in Power Plant Engineering By S.Domkundwar.
2. A Course in Power Plant Engineering By T. Morse.
3. A Course in Power Plant Engineering By Nagpal.
4. A Course in Power Plant Engineering By Agrawal.



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DIPLOMA IN MECHANICAL ENGINEERING

ME-6004 MAJOR PROJECT

SIXTH SEMESTER

S.NO	CONTENT
1	Design and Make Type Projects : Suggested Projects : Lab equipment to demonstrate and verify the principle of conservation of momentum either linear angular with an accuracy of 5%. A lathe tool dynamometer to measure cutting forces. A working model of a variable speed drive, which when given a fixed r.p.m. input, will be able to give a steeples variable output r.p.m., within the range of 3 times the in put to 1/3 times the input. A Sheet metal cutting machine, useful to village blacksmiths in preparing articles like hand pumps, funnels, containers etc. A Wood working machine, hand or pedal operated to be useful in a village situation. Equipment to provide hot water in your hostel, using solar energy. A suitable hand operated press for producing ornamental tiles. A device to utilize energy from the wind, for drawing water from a well for irrigating a small farm. A Design and development of hand operated grinder/juicer. Design and development of special furniture, e.g. for hospitals, handicapped persons etc. Material handling equipment to suit the needs of a local industry. Design and development of jigs / fixture for a live production situation.



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DIPLOMA IN MECHANICAL ENGINEERING

ME-6004 MAJOR PROJECT

SIXTH SEMESTER

S.NO	CONTENT
2	Investigation Projects : Suggested Themes : Selection of a suitable machine tool to produce a given component with specified tolerances, economically. Analyzing the causes and suggesting remedial steps for a machine which is functioning erratically. Causes of bearing failure in a given situation. Investigation in to the cutting forces set up on a lathe for different feeds, speeds and materials. Determining the most economical, running conditions for an. I.C. engine. Investigation in to the different methods of measurements of flow/ temperature/ pressure with a view to suggest a suitable method for a live situation. Installing, commissioning and fault rectification of a test/ measuring set up or a machine.
3	Feasibility Studies : Suggested Products : Packaging material. Household utensils. Furniture, Match boxes, Wire nails, Vulcanizing unit. Time Piece components, plastic buckets, Aluminum anodized titan boxes, Raxine bags, folders etc. Washing soap, Instrument boxes for schools.



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ME-6004 MAJOR PROJECT

SIXTH SEMESTER

LIST OF SUGGESTED SKILLS TO ACQUIRE

S.No.	SUGGESTED SKILLS
1	Participate effectively in group work.
2	Analyse, synthesise and evaluate at technician level while dealing with engineering situations.
3	Apply his knowledge in practical situations.
4	Be able to plan ahead.
5	Be able to take appropriate decisions.
6	Maintain good human relations, by possessing social skills and tolerance.
7	Be able to arrive at creative solutions to problems.
8	Demonstrate self reliance and self discipline.
9	Understand and accept his own strengths and limitations
10	Adapt readily to changing environments.
11	Have a sense of purpose and pride of achievement.
12	Demonstrate his initiative.
13	Reliably work independently.
14	Be able to seek, select, use and present information's.



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ME-6004 MAJOR PROJECT

SIXTH SEMESTER

S.No	The following points, concerning the project should be noted :
1	A record of all calculations, drawings and designs must be kept.
2	Student will work either individually or in a group of 2, 3, or
3	A Written report must be available to the supervisor at the end of the course. This report should be neatly written and produced in a suitable folder which bears the name of the polytechnic, the title of the project and the name of the contributor and the dates of the work. Although the reports will obviously vary from project to project, yet, they should in general, conform to the following pattern : (a) Summary: A summary of the report which should not exceed one page in length. (b) Index: The report should be logically indexed. (c) Introduction: This should introduce the reader to the objectives of the exercise. (d) Main body: this will vary considerably from project to project and will contain all design calculations, drawings, results etc. (e) Conclusion: This will state the main conclusions of the exercise. (f) Bibliography: A list of all references used. In general, you will find that a well presented brief concise and logical report will score a higher assessment than a badly presented, lengthy muddled and illogical report.



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ME-6004 MAJOR PROJECT

SIXTH SEMESTER

S.No.	The following points, concerning the project should be noted :
1	The examiners will look for the following points when assessing your project : (a) The way in which you have applied existing knowledge to your project. (b) Mental skill in manipulations of formulas etc. (c) The quality of physical skills in the manufacture and assembly of apparatus and test pieces etc. (d) Analysis of test results to produce a conclusion. (e) The ability to produce a complete project from all the individual elements. (f) Skill in evaluating between, for instance, various design alternatives. (g) The way in which the initial specifications are interpreted and translated in to a finished project. (h) The quality of the creative aspect of the design to meet the given specifications. (i) The quality of the written content of the report. (j) The quality to the graphical content of the report.



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The assessment of your project will count towards your final diploma, and the accompanying marking grid shows how the marks are allotted in your particular case.

Remember that this is your project. The supervisor is there to give guidance if and when it is necessary. He will start you off on the right lines but will only intervene at the later stages, if asked.



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DIPLOMA IN MECHANICAL ENGINEERING

ME-6005 PROFESSIONAL ACTIVITY

SIXTH SEMESTER

Topic No	Contents	Hours
01	Industrial Visits Structured industrial visits be arranged and report of the same shall be submitted by the individual student, to form a part of the term work.(2 visits) Following are the suggested types of Industries/ Fields - i) Automobile manufacturing / auto component manufacturing units to observe the working of SPM ii) Refrigeration and air conditioning manufacturing / servicing units / industries / workshops iii) Automobile service stations for four wheelers iv) Co-ordinate measuring machine to observe its construction working specifications and applications. v) Auto Engine Testing unit to gather details regarding the testing procedures/parameters etc. vi) Wheel Balancing unit for light and/or heavy motor vehicles. vii) Food processing unit. viii) Textile industry machinery manufacturing / servicing units. ix) Hydro electric and Thermal power plants. x) Engine testing, exhaust gas analysis and vehicle testing xi) PWD workshop.	09



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ME-6005 PROFESSIONAL ACTIVITY

SIXTH SEMESTER

Topic No	Contents	Hours
02	The Guest Lecture/s From field/industry experts, professionals to be arranged (2 Hrs duration), minimum 4 nos. from the following or alike topics. The brief report to be submitted on the guest lecture by each student as a part of Term work a) Electronic fuel injection systems b) Exhaust gas analysis. c) Vehicle testing. d) Transducer application in automobiles. e) Environmental pollution & control. f) Vehicle aerodynamics & design. g) Earth moving machines. h) Automobile pollution, norms of pollution control. i) Biotechnology j) Nanotechnology k) Rapid prototyping l) Programmable logic controllers m) TQM n) MPFI o) Hybrid motor vehicles p) Packaging technology q) Appropriate technology r) Six sigma systems s) LPG / CNG conversion kit.	08



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DIPLOMA IN MECHANICAL ENGINEERING

ME-6005 PROFESSIONAL ACTIVITY

SIXTH SEMESTER

Topic No	Contents	Hours
03	Group Discussion : The students should discuss in group of six to eight students and write a brief report on the same, as a part of term work. The topic of group discussions may be selected by the faculty members. Some of the suggested topics are (any one)- i) CNG versus LPG as a fuel. ii) Petrol versus Diesel as a fuel for cars. iii) Trends in automobile market. iv) Load shading and remedial measures. v) Rain water harvesting. vi) Trends in refrigeration Technology. vii) Disaster management. viii) Safety in day to day life. ix) Energy Saving in Institute. x) Nano technology.	12



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04	Seminar : (any 2 topics) Seminar topic should be related to the subjects of fifth semester / topics from guest lectures. Students shall submit a report of at least 10 pages and deliver a seminar (Presentation time – 10 minutes for a group of 2 students)	12
05	Individual Assignments : Any two from the list suggested a) Process sequence of any two machine components. b) Write material specifications for any two composite jobs. c) Collection of samples of different plastic material or cutting tools with properties , specifications and applications. d) Preparing models using development of surfaces. e) Assignments on bending moment , sheer forces , deflection of beams and torsion chapters of strength of material. f) Select different materials with specifications for at least 10 different machine components and list the important material properties desirable. g) Select 5 different carbon steels and alloy steels used in mechanical engineering applications and specify heat treatment processes employed for improving the properties. Also give brief description of the heat treatment processes. h) List the various properties and applications of following materials – a.Ceramics, b. fiber reinforcement plastics,c. thermo plastic plastics d. thermo setting plastics,e. rubbers. OR Conduct ANY ONE of the following activities through active participation of students and write report i) Rally for energy conservation / tree plantation. ii) Survey for local social problems such as mal nutrition, unemployment,cleanliness, illiteracy etc.	08



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	iii) Conduct aptitude , general knowledge test , IQ test iv) Arrange any one training in the following areas : a) Yoga. B) Use of fire fighting equipment and First aid Maintenance of Domestic appliances.	
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ME-6005 PROFESSIONAL ACTIVITY

SIXTH SEMESTER

07	Student Activities – Students in a group of 3 to 4 shall perform ANY TWO of the following activities (Other similar activities may be considered) and write a report as a part of term work. Activities :- 1. Collection of data regarding loan facilities or other facilities available through different organizations / banks to budding entrepreneurs 2. Survey and interviews of successful entrepreneurs in near by areas 3. Survey of opportunities available in thrust areas identified by Government or DIC. 4. Measuring Screw thread parameters on floating carriage dial micrometer and select the optimum diameter of wire. 5. Survey of data regarding different types of pumps with specifications from manufacturers catalogue, local markets, end users (any other engineering products may be considered for survey) 6. Survey of farm implements used by farmers	05
	Total	60



BHABHA UNIVERSITY, BHOPAL (M.P.)

DIPLOMA IN MECHANICAL ENGINEERING

ME-6005 PROFESSIONAL ACTIVITY

SIXTH SEMESTER

Learning Resources:

Books:

Sr. No	Author	Title of the book	Publisher
1	Mark Ratner and Daniel Ratner	Nanotechnology	Pearson Education, New Delhi
2	Yoram Korem	Computer Control of Manufacturing System	Mcgraw Hill Publication
3	Sunil Chopra, Peter Meindl	Supply Chain Management	Pearson Education, New